development of housing pressure and residential vacancy in Dutch municipalities between 2020 and 2025, and understanding coexistence and scarcity through price pressure, rental structure and urban density.

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Part 1 - Identify a Social Problem

1.1 Describe the Social Problem

The Dutch housing market is under unprecedented pressure. Across the Netherlands, rental prices have risen sharply, buying a home has become unaffordable for many households, and waiting lists for social housing continue to grow (Centraal Bureau voor de Statistiek [CBS], 2025; Kences, 2025). At the same time, a paradox emerges: despite the apparent shortage, a significant number of dwellings remain unoccupied in many municipalities.

This combination of housing pressure and vacancy creates a social problem that does not affect all of the Netherlands equally. In some municipalities, housing pressure is severe and vacancy is low; in others, vacancy rates are surprisingly high despite national shortages. This spatial variation means that the opportunity to find affordable housing increasingly depends on where one lives a form of geographic inequality that affects young adults, lower income households, and people seeking to relocate.

Why is this relevant?

This problem is relevant for several reasons:

1. Economic relevance. The housing market is a cornerstone of the economy. Rising housing costs reduce disposable income, limit labor mobility, and contribute to wealth inequality between homeowners and renters (CBS, 2025). Geographic differences in housing pressure also shape patterns of regional economic development.

2. Social relevance. Access to affordable housing is widely considered a basic right. When access depends heavily on geography, spatial inequality deepens and social cohesion is undermined. The coexistence of shortage and vacancy raises difficult questions about the efficiency and fairness of the housing system.

3. Policy relevance. Understanding where housing pressure is most severe and where it coincides with vacancy helps policymakers target interventions more effectively. A municipality-level analysis can inform decisions about new construction, rent regulation, and vacancy policies, and shows which areas need attention most urgently.

4. Personal relevance. As students at VU Amsterdam, we experience the consequences of housing pressure directly. By analyzing the problem at a national scale, we move beyond our own situation and investigate the structural forces shaping housing access across the Netherlands.

By quantifying and visualizing how housing pressure and vacancy vary across Dutch municipalities between 2020 and 2025, this project aims to make the scale and distribution of the problem visible, and to identify which types of municipalities face the sharpest paradox between scarcity and unused capacity.

Part 2 - Data Sourcing

2.1 Load in the data

The data were obtained from the CBS publication *Kerncijfers wijken en buurten*. The Excel files for the years 2020–2025 were downloaded from the CBS website and stored locally outside the GitHub repository. The files were then imported and combined into a single dataset for analysis. The data were obtained from the Centraal Bureau voor de Statistiek (2025). Each year is provided as a separate dataset, resulting in six datasets in total, which were merged into one combined panel dataset covering 2020–2025.

```
mdf <- read.csv(here("data", "clean", "CLEANED_municipal_housing_2020_2025.csv"))
```

2.2 Provide a short summary of the datasets

```
head(mdf)
```

```
##   year gwb_code     gm_naam     recs a_woning   a_hh p_leegsw g_wozbag p_koopw
## 1 2020   GM0003  Appingedam Gemeente     6042   5528        7   155000      48
## 2 2020   GM0010    Delfzijl Gemeente    12223  11591        7   138000      61
## 3 2020   GM0014   Groningen Gemeente   116445 136383        4   208000      41
## 4 2020   GM0024   Loppersum Gemeente     4687   4297        8   175000      65
## 5 2020   GM0034      Almere Gemeente    85977  89319        2   251000      64
## 6 2020   GM0037 Stadskanaal Gemeente    15292  14594        5   175000      59
##   p_huurw p_wcorpw p_ov_hw ste_mvs bev_dich  a_inw housing_pressure
## 1      52       43       9       3      490  11642        0.9149288
## 2      39       28      11       4      185  24678        0.9482942
## 3      59       33      25       1     1255 232874        1.1712225
## 4      35       28       7       5       86   9537        0.9167911
## 5      36       27       9       2     1640 211893        1.0388709
## 6      39       31       9       4      269  31686        0.9543552
##   effective_housing_stock vacancy_adjusted_pressure rental_vulnerability
## 1                 5619.06                 0.9837944                    9
## 2                11367.39                 1.0196712                   11
## 3               111787.20                 1.2200234                   26
## 4                 4312.04                 0.9965121                    7
## 5                84257.46                 1.0600723                    9
## 6                14527.40                 1.0045844                    8
##   paradox_index      urbanity_label price_pressure
```

```
## 1      6.404502    Moderately urban      0.4668675
## 2      6.638059      Slightly urban      0.4156627
## 3      4.684890 Very strongly urban      0.6265060
## 4      7.334329           Non-urban      0.5271084
## 5      2.077742      Strongly urban      0.7560241
## 6      4.771776      Slightly urban      0.5271084
```

This dataset comes from CBS and contains information about Dutch municipalities for the years 2020 to 2025. It includes data on topics such as population, housing, income, and households.

For our project, we mainly use the housing-related information. The dataset contains variables that can help us study housing pressure and vacancy across municipalities. Because the data cover multiple years, it is possible to see whether these patterns have changed over time.

The exact meaning and units of the variables differ and are explained in the CBS metadata (CBS, 2025) that accompany the dataset.

2.3 Describe the type of variables included

The dataset contains several types of variables relevant to our analysis. The core housing variables include `a_woning` (total housing stock) and `a_hh` (number of households). Vacancy is captured through `p_leegsw` (vacancy rate as a percentage of total housing stock).

Ownership and rental structure is captured through `p_koopw` (owner-occupied share), `p_huurw` (total rental share), `p_wcorpw` (corporation-owned rental share), and `p_ov_hw` (private rental share) — all expressed as percentages of total housing stock.

Urban density is measured through `ste_mvs` (urbanity code, ranging from 1 = very strongly urban to 5 = non-urban) and `bev_dich` (population density per km^2). Property values are captured through `g_wozbag` (average WOZ value in euros), which serves as a proxy for housing market prices.

All variables are measured at municipality level and are derived from administrative sources rather than surveys, which ensures high coverage but may mask within-municipality variation. The dataset does not include individual-level data, meaning inequality within municipalities cannot be observed directly.

Part 3 - Quantifying

3.1 Data cleaning

The raw CBS Excel files were cleaned using a separate R script (scripts/Datacleaning.R). The cleaning process involved several deliberate choices.

First, we filtered the dataset to municipality-level observations only (recs == "Gemeente"), excluding neighbourhood and district-level rows. This choice was made because municipalities are the most relevant unit for housing policy analysis they are the administrative bodies responsible for zoning, construction permits, and social housing allocation. Aggregating to a higher level would obscure important local variation, while neighbourhood-level data would introduce noise and missing values.

Second, we selected 15 variables from the original dataset of over 100 columns. The selected variables cover housing stock, vacancy, ownership structure, urbanity, population density, and property values — the core dimensions needed to quantify housing pressure and the housing paradox. Variables unrelated to housing (such as health or education indicators) were excluded to keep the analysis focused.

Third, the WOZ value variable (`g_wozbag`) was multiplied by 1000 to convert it from thousands of euros to euros, ensuring the unit is interpretable and consistent with other monetary variables.

One limitation of this cleaning approach is that municipalities change over time due to mergers. Some municipalities present in 2020 may not exist under the same name in 2025, which could introduce small inconsistencies in the longitudinal analysis.

3.2 Generate necessary variables

Variable 1: Housing Pressure

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.7942  0.9765  0.9898  0.9941  1.0048  1.3379
```

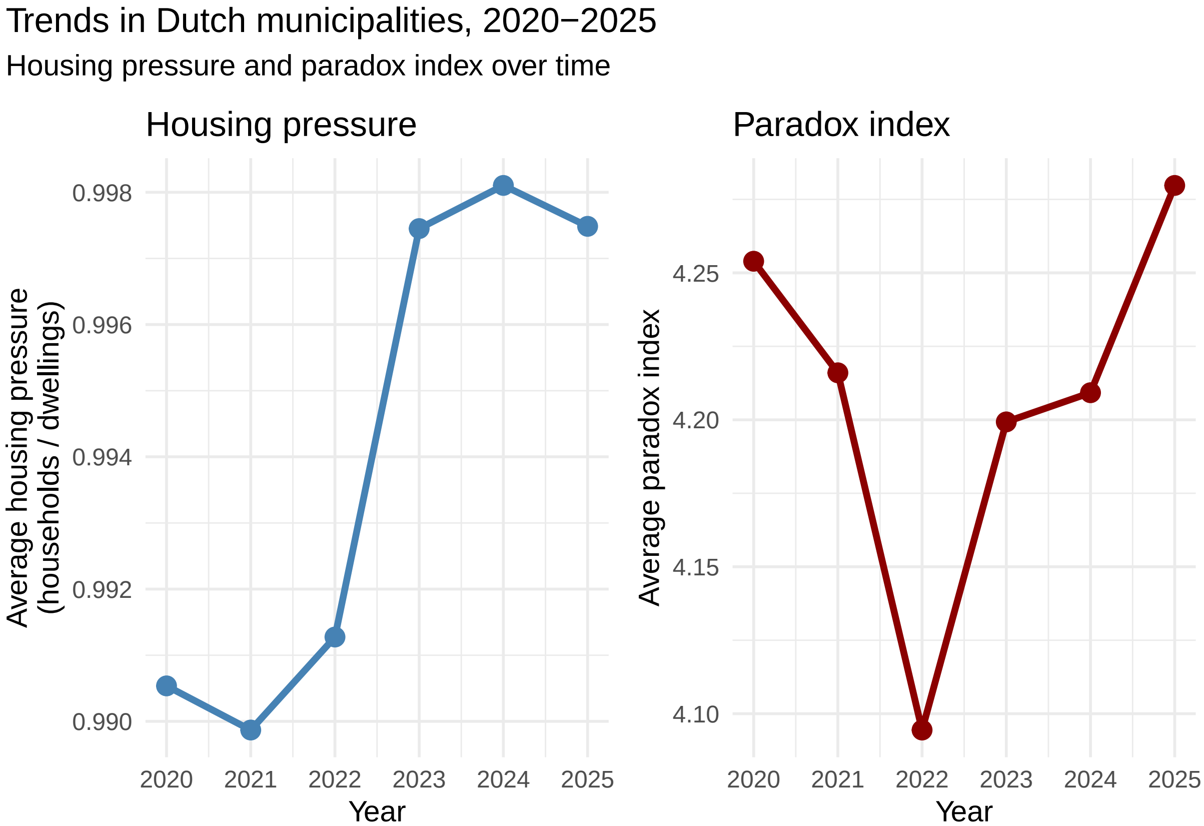
Housing pressure measures the ratio of households to dwellings in each municipality. A value above 1 indicates that there are more households than available dwellings, suggesting that demand exceeds supply.

Variable 2: Paradox Index

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.9967  2.9877  3.9157  4.2089  4.8032 21.9966
```

The paradox index combines housing pressure with vacancy to capture the central puzzle of our research: how can shortage and vacancy coexist? A high paradox index identifies municipalities with both high pressure and high vacancy the exact contradiction this project seeks to explain. Multiplying both values gives weight to municipalities where both problems occur simultaneously.

3.3 Visualize temporal variation

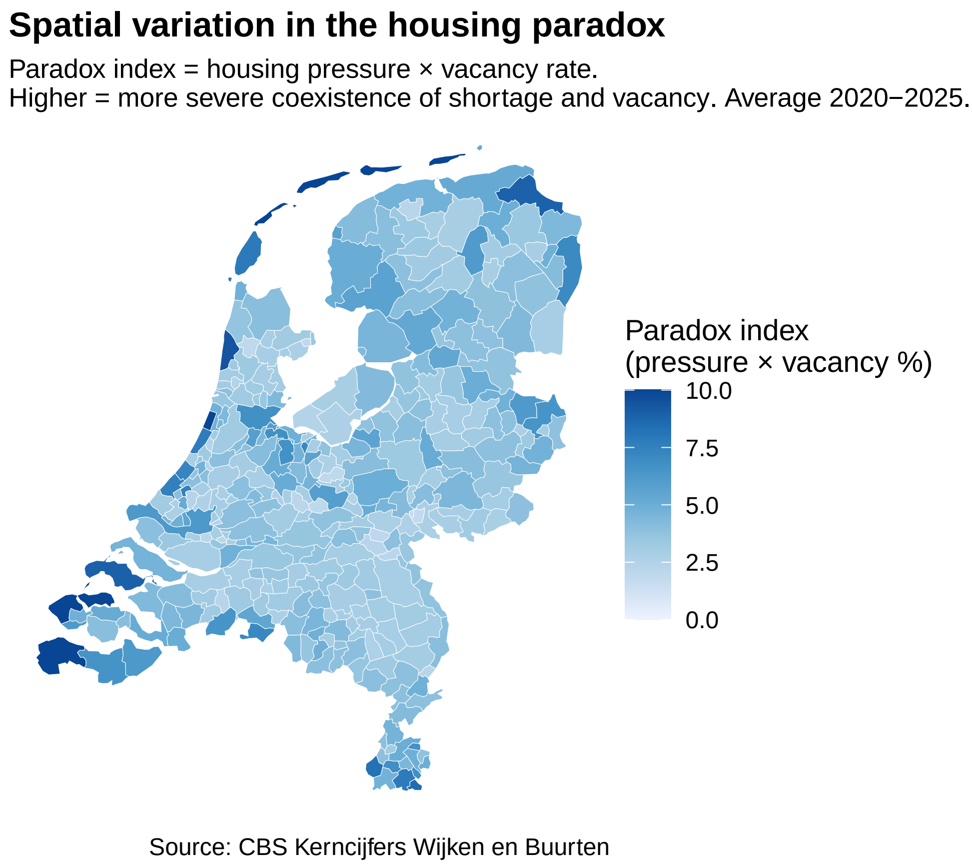


This figure shows the evolution of housing pressure and the paradox index across Dutch municipalities between 2020 and 2025. Each indicator is shown on its own scale to make changes over time visible.

Housing pressure (left panel) shows a clear upward trend: the average ratio of households to dwellings increased from approximately 0.990 in 2021 to 0.998 in 2024, before slightly decreasing in 2025. This pattern suggests that demand for housing has been rising faster than supply.

The paradox index (right panel) shows a more complex pattern: a sharp decline between 2020 and 2022, followed by a steady rise. The 2022 turning point may reflect post-pandemic housing market shifts, such as rising interest rates and changes in rental market policy. The resurgence of the paradox index in recent years suggests that shortage and vacancy are increasingly occurring side by side.

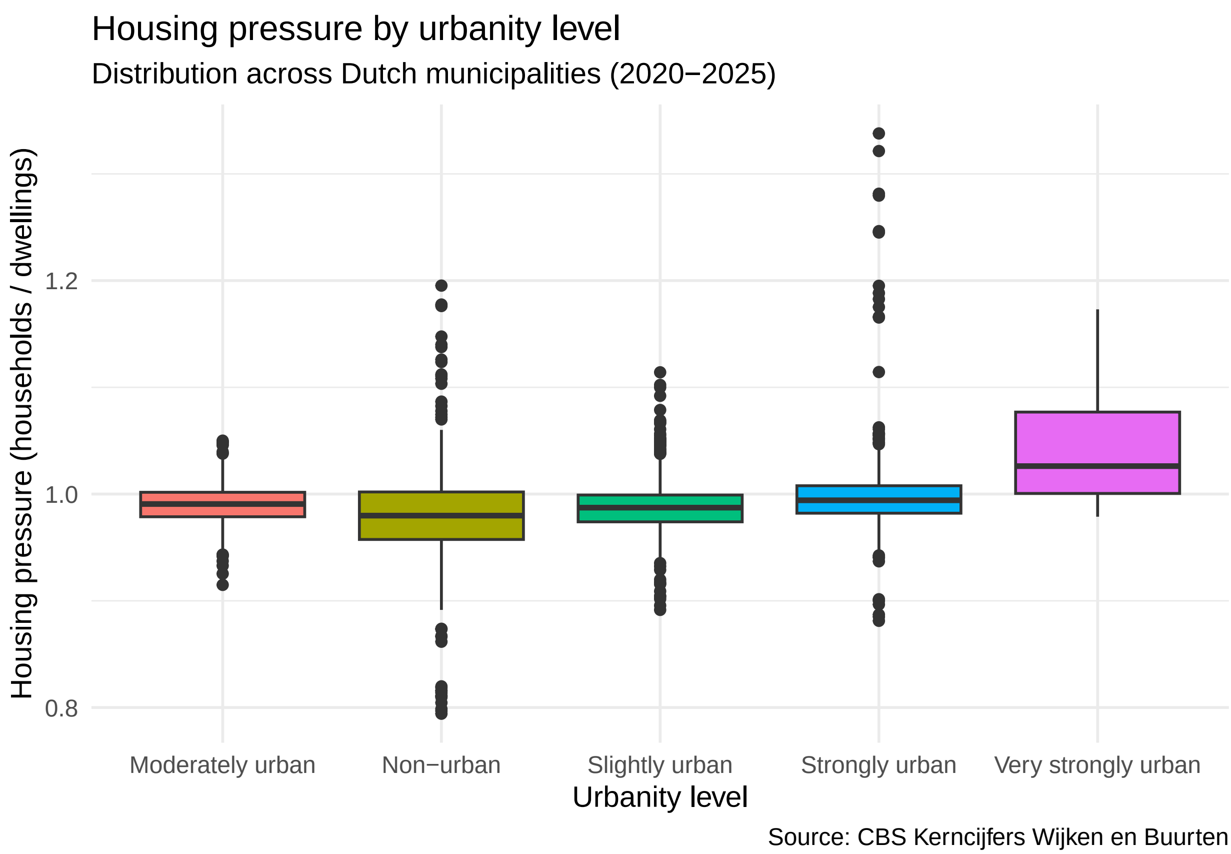
3.4 Visualize spatial variation



This map reveals substantial spatial variation in the housing paradox across Dutch municipalities. The highest paradox index values are concentrated in municipalities in the western Randstad region particularly around Amsterdam, Utrecht and The Hague as well as in parts of Noord-Brabant. This suggests that housing pressure and vacancy coexist most severely in already densely populated urban areas. Notably, several municipalities in Zeeland and Friesland also show elevated values, indicating that the paradox is not exclusively an urban phenomenon. This spatial heterogeneity underlines that national housing policy may be insufficient targeted regional interventions are likely needed to address the paradox where it is most acute.

3.5 Visualize sub-population variation

What is the distribution of housing pressure across municipalities with different urbanity levels?



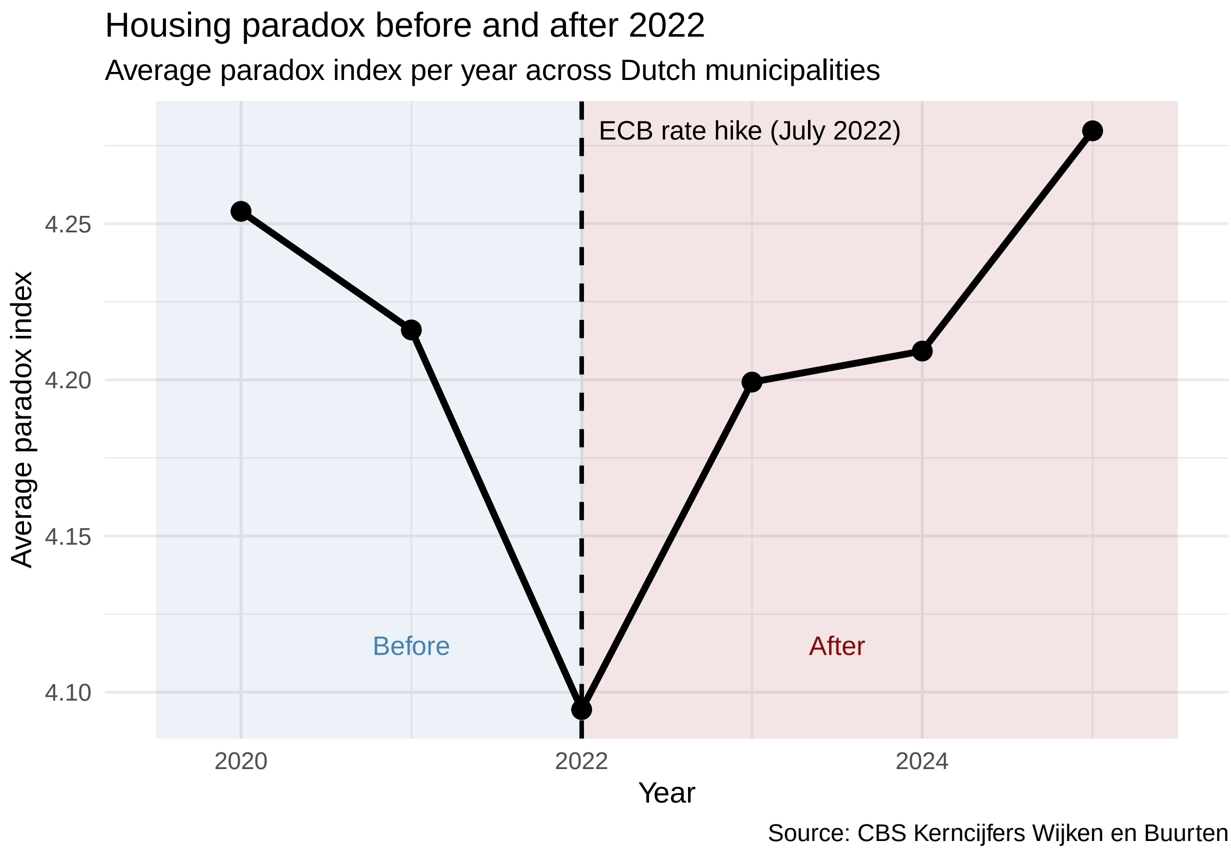
The figure shows that housing pressure is highest in strongly and very strongly urban municipalities, with median values close to 1.0 meaning nearly all dwellings are occupied. Non-urban and slightly urban municipalities show considerably more variation and lower median values, suggesting that housing markets in rural areas are less tight overall. The narrower interquartile range in urban categories indicates that high housing pressure is consistently present across urban municipalities, while rural areas show more heterogeneity. This pattern supports the hypothesis that the housing paradox is primarily driven by urban demand exceeding supply, and suggests that policy interventions targeting vacancy should focus on urbanised areas where pressure is most persistent.

```
## # A tibble: 5 x 4
##   urbanity_label      median_pressure mean_pressure n_municipalities
##   <chr>                         <dbl>         <dbl>            <int>
## 1 Very strongly urban           1.03          1.05                24
## 2 Strongly urban                0.994         1.00                80
## 3 Moderately urban              0.991         0.991               86
## 4 Slightly urban                0.987         0.988              138
## 5 Non-urban                     0.98          0.981               56
```

The table confirms the pattern shown in the boxplot. Very strongly urban municipalities show the highest median housing pressure (approximately 1.03), while non-urban municipalities have the lowest (approximately 0.98). This difference of roughly 0.05 points may appear small in absolute terms, but given that a value of 1.0 represents full occupancy, even small differences near the ceiling indicate meaningfully tighter housing markets in urban areas.

3.6 Event analysis

How did the housing paradox index change before and after the ECB interest rate hike in 2022? When the ECB raises interest rates, mortgage rates follow. Higher mortgage rates make borrowing more expensive, which reduces the number of housing transactions fewer people can afford to buy or move. This can simultaneously increase effective vacancy (owners unable to sell, properties sitting empty) while housing pressure remains high due to persistent demand. The 2022 rate hike is therefore a relevant external shock to test in the context of the housing paradox.



The figure shows that the housing paradox index declined slightly between 2020 and 2022, before rising sharply in the years following the ECB interest rate hike in July 2022 (Euronews, 2022). By 2025, the paradox index reached its highest recorded value in the dataset, suggesting that rising borrowing costs may have simultaneously suppressed housing transactions increasing effective vacancy while structural housing shortages persisted. This pattern is consistent with the hypothesis that higher mortgage rates reduced housing market liquidity, trapping both buyers and sellers and thereby intensifying the coexistence of pressure and vacancy. However, this association does not establish causality: other concurrent developments such as construction delays, demographic shifts, or changes in rental market regulation may also have contributed to the observed trend. A more rigorous causal analysis would require controlling for these confounding factors.

Part 4 - Discussion

4.1 Discuss your findings

Our analysis of Dutch municipalities between 2020 and 2025 reveals three key findings about the housing paradox.

First, housing pressure has increased consistently over the study period, with the average ratio of households to dwellings rising from approximately 0.990 in 2020 to 0.998 in 2024. This confirms the pattern identified in the literature: structural housing shortages are worsening as demand continues to outpace supply (CBS, 2025; Ministerie van Binnenlandse Zaken en Koninkrijksrelaties, 2024).

Second, the paradox index which captures the simultaneous occurrence of housing pressure and vacancy shows a V-shaped pattern over time. The decline between 2020 and 2022 may reflect pandemic-related reductions in mobility and vacancy, while the sharp rise after 2022 coincides with the ECB interest rate hike. Rising mortgage rates appear to have frozen the housing market: fewer transactions meant that both buyers and sellers became stuck, increasing effective vacancy while pressure remained high. This is consistent with the findings of Planbureau voor de Leefomgeving (2024), which identified liquidity constraints as a key driver of spatial housing mismatches.

Third, the spatial and subgroup analyses reveal that the paradox is not uniform. Urban municipalities face the highest and most consistent housing pressure, while the paradox index is elevated in both Randstad municipalities and some peripheral areas such as Zeeland and Friesland. This suggests that the causes of the paradox differ by region: in urban areas it is driven by demand, while in peripheral areas vacancy may reflect demographic decline.

Limitations. Several limitations should be acknowledged. The paradox index is a simple multiplicative measure and does not account for the types of vacant dwellings or the reasons for vacancy. Municipality mergers over the study period may introduce minor inconsistencies. Furthermore, all analyses are observational the association between the ECB rate hike and the paradox index does not establish causality, and confounding factors such as construction delays and demographic shifts cannot be ruled out.

Part 5 - Reproducibility

5.1 Github repository link

https://github.com/Marlinehuman/ProgrammingForEconG5

5.2 Reference list

Centraal Bureau voor de Statistiek. (2025). *Kerncijfers wijken en buurten 2025* [Dataset]. CBS. https://www.cbs.nl/nl-nl/reeksen/kerncijfers-wijken-en-buurten

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